

DART: Distributed 4-AFSK Radio Telemetry

Physical Layer Specification & Demodulation Architecture

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Abstract

This paper presents the formal specification and signal-processing architecture of the **Distributed 4-AFSK Radio Telemetry (DART)** system. In network protocol engineering, beginners frequently conflate telemetry software with application-layer network daemons. DART operates strictly as an **OSI Layer 1 (Physical / PHY)** acoustic-frequency-shift-keying protocol and baseband framing standard designed for standard voice-grade amateur transceiver channels (audio passband 300–3000 Hz). This paper decomposes the mathematical basis of Gray-coded 4-AFSK tone plans, discrete-time coherent tone correlation, non-data-aided early/late symbol timing recovery, matrix interleaving, $r = 1/2$ constraint-length $K = 7$ Viterbi convolutional soft-decision forward error correction (FEC), structural squelch detection, and time-diversity chase combining. Furthermore, it details the integrated cryptographic framing subsystem, providing authenticated encryption with associated data (AEAD) via Argon2id key derivation, XChaCha20-Poly1305 sealing, BLAKE2b key fingerprint identification, memory zeroization, and physical data shredding.

1 Introduction & ISO/OSI Reference Mapping

A prevalent misconception in modern embedded systems is treating telemetry as a high-level software application (such as an MQTT broker or JSON over HTTP). In wireless communications, robust long-range transmission begins at **Layer 1: The Physical Layer (PHY)** of the ISO/OSI model. Without a deterministic physical modulation, synchronisation, and error-correction standard, higher layers cannot function across unreliable radio frequency (RF) channels.

ISO Layer	Function	DART Protocol Mapping
Layer 7–4	Application / Session	User Payload (Telemetry sensor records, ASCII packets)
Layer 3	Network Layer	Node Addressing (Configurable 3-Byte Hexadecimal ID: nodeid)
Layer 2	Data Link Layer	Framing, CRC-8 (Header), CRC-16 (Payload), Matrix Block Interleaver
Layer 1.5	Cryptographic Sublayer	Argon2id KDF, XChaCha20-Poly1305 AEAD, BLAKE2b Key ID
Layer 1	Physical (PHY)	4-AFSK Baseband Modulator / Demodulator, Bit Demapper, Tone Correlator, Timing Error Detector (TED), Squelch / PTT Control

Figure 1: ISO/OSI reference model alignment for DART.

The **Distributed 4-AFSK Radio Telemetry (DART)** standard was engineered to transmit digital data through standard, un-modified analog frequency modulation (FM) and single-sideband (SSB) transceivers using baseband Audio Frequency Shift Keying (AFSK). The design targets hostile RF propagation scenarios: mobile multipath fading, selective channel attenuation, audio flutter, and low signal-to-noise ratio (SNR) voice-bandwidth lines ($B \approx 2.7$ kHz).

2 Modulation Theory: 4-AFSK Tone Planning

DART maps binary sequences to discrete audible tones transmitted over a baseband audio channel sampled at $F_s = 48,000$ Hz.

2.1 Tone Synthesis and Orthogonality

In classic Bell 202 2-AFSK (used in standard AX.25 APRS), two audio tones (1200 Hz Mark, 2400 Hz Space) represent binary 1 and 0 at a rate of 1200 baud, resulting in 1200 bps throughput with poor spectral efficiency (0.5 bps/Hz). DART implements **M-ary Frequency Shift Keying with $M = 4$ (4-AFSK)**, packing two bits (a *dibit*) into each transmitted symbol period $T_{\text{sym}} = 1/R_{\text{baud}}$.

The tone plan is centered at $f_c = 1700$ Hz. The frequency allocated to tone index $k \in \{0, 1, 2, 3\}$ is defined as:

$$f_k = f_c + \left(k - \frac{M-1}{2}\right) R_{\text{baud}} = 1700 + (k - 1.5) \cdot R_{\text{baud}} \quad (1)$$

At nominal $R_{\text{baud}} = 300$ baud, the tone frequencies evaluate to:

- **Tone 0:** $1700 + (-1.5 \times 300) = 1250$ Hz
- **Tone 1:** $1700 + (-0.5 \times 300) = 1550$ Hz
- **Tone 2:** $1700 + (+0.5 \times 300) = 1850$ Hz
- **Tone 3:** $1700 + (+1.5 \times 300) = 2150$ Hz

Because adjacent tones are spaced precisely at $\Delta f = R_{\text{baud}} = 300$ Hz, the tones are **mutually orthogonal** over the symbol duration:

$$\int_0^{T_{\text{sym}}} \cos(2\pi f_i t + \phi_i) \cos(2\pi f_j t + \phi_j) dt = 0, \quad \forall i \neq j \quad (2)$$

This allows non-coherent quadrature cross-correlation at the demodulator without inter-carrier interference (ICI). All tones comfortably occupy the amateur transceiver passband (300–3000 Hz).

2.2 Gray Mapping

To minimize the Bit Error Rate (BER), dibits are mapped to tone frequencies using a reflected Gray code. In fading environments, adjacent-frequency classification errors are the dominant failure mode. Gray coding guarantees that selecting an adjacent frequency bin causes only a single bit error:

Tone Index (k)	Frequency (Hz)	Gray Dibit ($b_1 b_0$)
0	1250	00
1	1550	01
2	1850	11
3	2150	10

3 Physical Framing, Synchronisation & Forward Error Correction

Raw RF bitstreams are subject to noise bursts, radio squelch tails, and bit slips. DART addresses these using a structured physical frame architecture.

3.1 Frame Geometry

A DART transmission begins with an unmodulated carrier burst, followed by synchronization words and an FEC-protected payload structure:

Carrier Preamble	Bit Preamble	ASM (Sync Word)	Length Header	Interleaved Body
1000 ms Tone	128 Bits (64 Syms)	32 Bits (16 Syms)	60 Coded Bits	$R \times C$ Coded Bits

Figure 2: DART Physical Layer Frame Layout on the air interface.

1. **Carrier Preamble:** A single unmodulated tone ($f_0 = 1250$ Hz) lasting up to 1000 ms to wake up analog hardware squelch and allow audio input auto-gain control (AGC) circuits to stabilize.
2. **Bit Preamble (128 bits):** Alternating outer tones (1250 Hz $\leftrightarrow$ 2150 Hz, dibits 00 $\leftrightarrow$ 10) generating maximum frequency deviation transitions ($\Delta f = 900$ Hz) every symbol to drive clock recovery lock.
3. **Attached Sync Marker (ASM):** A 32-bit unique correlation sequence:

$$\text{SYNC_WORD} = 0x1ACFFC1D \quad (\text{Binary: } 0001\ 1010\ 1100\ 1111\ 1111\ 1100\ 0001\ 1101_2)$$

This sequence provides ideal aperiodic auto-correlation properties, tolerating up to 3 corrupted bits during correlation search.

4. **Frame Header (60 coded bits):** Encodes a 16-bit payload length field followed by an 8-bit CRC ($g(x) = x^8 + x^2 + x + 1$). The resulting 24 bits are appended with 6 zero-flush bits and encoded with a rate-1/2 convolutional code $(24 + 6) \times 2 = 60$ coded bits.
5. **Payload Body:** Encapsulates raw data (ASCII telemetry, binary sensors, or authenticated ciphertext) protected by a 16-bit CRC-CCITT ($g(x) = x^{16} + x^{12} + x^5 + 1$).

3.2 Convolutional Forward Error Correction

DART uses the deep-space standard NASA planetary convolutional code:

- **Code Rate (r):** 1/2 (2 coded bits emitted per 1 information bit).
- **Constraint Length (K):** 7 (Memory $m = K - 1 = 6$ shift register delay stages, 64 trellis states).
- **Generator Polynomials:**

$$G_1 = 1111001_2 = 171_8 = 0x79 \quad (3)$$

$$G_2 = 1011011_2 = 133_8 = 0x5B \quad (4)$$

At each bit step t , with input bit u_t and current state register $S_t = (s_5, s_4, s_3, s_2, s_1, s_0)$, the parity output bits are:

$$c_1 = u_t \oplus s_5 \oplus s_4 \oplus s_3 \oplus s_2 \oplus s_0, \quad c_2 = u_t \oplus s_5 \oplus s_3 \oplus s_2 \oplus s_0 \quad (5)$$

Six zero-tail flush bits ($u = 0$) are shifted through the encoder at the end of the block to return the trellis to the known all-zero state ($S_0 = 0$).

3.3 Matrix Block Interleaver

Radio channels frequently experience burst noise (e.g., ignition interference, atmospheric static, and RF carrier drops) that corrupts consecutive symbols, overwhelming the Viterbi decoder's constraint window ($K = 7$). DART prevents this via an adaptive square/rectangular matrix block interleaver.

Given N_{coded} bits, the interleaver derives its dimensions dynamically:

$$R = \lfloor \sqrt{N_{\text{coded}}} \rfloor, \quad C = \left\lceil \frac{N_{\text{coded}}}{R} \right\rceil \quad (6)$$

Bits are written row-wise into an $R \times C$ matrix and read out column-wise for RF transmission. A burst of length $L \leq C$ on the RF link is deinterleaved on receive into isolated single-bit errors separated by R bits, well within the Viterbi engine's error-correction capacity.

4 Demodulation & Signal Processing Architecture

The receiver (dartRX) transforms continuous PCM audio from an audio device into verified telemetry frames through several DSP stages.

Audio In	RMS Power Squelch	Quadrature Correlator	Early-Late TED	Soft Demapper
48 kHz PCM	Threshold ≥ -30 dBFS	I/Q Energy per Tone	Symbol Centering	Metric Weighting

Figure 3: Baseband DSP receiver pipeline.

4.1 Continuous Squelch & Capture Windowing

The receiver computes the root-mean-square (RMS) energy in decibels relative to full scale (dBFS):

$$\text{dBFS} = 20 \log_{10} \left(\frac{\sqrt{\frac{1}{N} \sum_{n=0}^{N-1} x[n]^2}}{32768} \right) \quad (7)$$

When energy exceeds the threshold (typically -30.0 dBFS), recording triggers. To ensure transmission ends cleanly without clipping the frame tail, a hang timer keeps the buffer open for up to 2000 ms after signal drop.

4.2 Discrete Quadrature Tone Correlation

For each symbol evaluation window starting at sample index p , the energy for each of the $M = 4$ candidate tones is extracted using matched quadrature filters:

$$I_k = \sum_{s=0}^{L_{\text{integ}}-1} x[p+s] \cdot \cos \left(2\pi f_k \frac{s}{F_s} \right) \quad (8)$$

$$Q_k = \sum_{s=0}^{L_{\text{integ}}-1} x[p+s] \cdot \sin \left(2\pi f_k \frac{s}{F_s} \right) \quad (9)$$

$$E_k = I_k^2 + Q_k^2 \quad (10)$$

The integration window length is selected as $L_{\text{integ}} = \lfloor 0.9 \cdot (F_s / R_{\text{baud}}) \rfloor = 144$ samples at 300 baud. This avoids evaluating symbol transition boundaries while maintaining high frequency selectivity.

4.3 Timing Synchronization: Phase Search and Early-Late Gate

Symbol timing recovery is achieved in two stages:

1. **Coarse Phase Acquisition:** The detector evaluates 16 fractional-phase offsets within a symbol period ($\Delta\tau = \frac{T_{\text{sym}}}{16}$) searching for the 32-bit ASM sync word. Once found, this establishes grid-phase lock.
2. **Early-Late Gate Tracking:** To combat clock drift between disparate sound cards, an Early-Late Timing Error Detector (TED) continuously inspects tone energy at offsets $\pm\tau_{\text{gate}}$ ($\tau_{\text{gate}} = \frac{1}{4}T_{\text{sym}}$). If late energy exceeds early energy ($E_{\text{Late}} > 1.01 \cdot E_{\text{Early}}$), the sampling pointer advances by +0.5 samples; conversely, it retards by -0.5 samples when early energy dominates.

4.4 Soft-Decision Metric Generation and Viterbi Decoding

Rather than making hard binary decisions (0 or 1), DART calculates continuous-valued soft metrics based on the relative tone energies:

$$\text{Confidence} = \frac{|E_{\text{dominant}} - E_{\text{secondary}}|}{E_{\text{dominant}} + E_{\text{secondary}}} \times \min\left(1.0, \frac{E_{\text{dominant}}}{E_{\text{ref}}}\right) \quad (11)$$

If the confidence falls below an erasure floor ($\gamma_{\text{floor}} = 0.15$), the bit metric is zeroed (erasure). The soft metrics feed into a 64-state soft-input Viterbi decoder, yielding up to 5.2 dB of coding gain over unencoded FSK in additive white Gaussian noise (AWGN) channels.

4.5 Chase Combining Time Diversity

If configured for repeat transmissions ($N_{\text{repeats}} \geq 2$), DART does not simply discard frames with failing CRC checks. It stores the unquantized soft metrics $\lambda_i^{(c)}$ of each repeat copy c and performs equal-gain packet combining:

$$\Lambda_i = \sum_{c=1}^{N_{\text{copies}}} \lambda_i^{(c)} \quad (12)$$

Passing the synthesized super-frame Λ back into the Viterbi trellis enables error-free decoding from multiple transmissions that individually failed CRC validation.

5 Cryptographic Subsystem & Security Architecture

Operating at the physical/link boundary requires that security primitives be lightweight, constant-time, and resilient against over-the-air bit flips. DART integrates modern authenticated encryption with associated data (AEAD) to guarantee payload confidentiality, authenticity, and cryptographic integrity.

5.1 Key Derivation via Argon2id

Shared passkeys configured on field nodes are hardened against offline brute-force and GPU/ASIC dictionary attacks using **Argon2id** (the winner of the Password Hashing Competition), combining data-dependent and data-independent memory access passes:

- **Memory Cost (m):** Configurable from 8 MiB to 1024 MiB (default: 64 MiB work memory).
- **Time Cost (t):** 3 iterations.

- **Parallelism (p):** 1 lane / 1 thread (tailored for low-power embedded single-board computers).
- **Salt (S):** A 16-byte (32 hexadecimal characters) globally synchronized system salt (`kdf_salt`) shared across the network to prevent multi-target table precomputation.

Argon2id produces a high-entropy 256-bit symmetric key $K \in \{0, 1\}^{256}$.

5.2 XChaCha20-Poly1305 AEAD Wire Framing

Payload confidentiality and tamper resistance are provided using **XChaCha20-Poly1305**. When encryption is active, the raw payload is transformed into an AEAD envelope:

Random Nonce	Poly1305 MAC Tag	Encrypted Ciphertext
24 Bytes (192 bits)	16 Bytes (128 bits)	N_{bytes} (Max 472 Bytes Plaintext)

1. **Extended Nonce (24 Bytes):** Standard ChaCha20 uses a 96-bit nonce, which carries birthday-bound collision risks when keys are reused across thousands of automated telemetry broadcasts. DART uses the extended 192-bit nonce format (XChaCha20), where nonces are drawn directly from kernel CSPRNG entropy (`getentropy` or `/dev/urandom`). The collision probability across trillions of frames remains negligible.
2. **Authentication Tag (16 Bytes):** A Poly1305 one-time authenticator verifies that neither the ciphertext nor the 24-byte nonce has suffered RF corruption or malicious injection.
3. **Overhead Accounting:** The combined AEAD envelope adds exactly 40 bytes of overhead (24 + 16). With a physical transmission cap of 512 bytes, the maximum plaintext telemetry record is bounded to $512 - 40 = 472$ bytes.

5.3 Domain-Separated Key Identification (BLAKE2b Fingerprint)

To permit operators to diagnose key mismatches over radio networks without exposing secrets, DART computes a 6-character hexadecimal Key ID using a domain-separated BLAKE2b keyed hash:

$$\text{KeyID} = \text{HexEncode}\left(\text{BLAKE2b}(\text{Key} = K, \text{Msg} = \text{"DART-keyid-v1"})[0..2]\right) \quad (13)$$

Transmitters log their Key ID alongside outgoing frames; receivers logging an authentication failure inspect the active Key ID to determine if a peer node is operating with an un-updated passkey or incorrect salt.

5.4 Memory Zeroization & Physical File Shredding

In remote telemetry stations (e.g., solar-powered repeaters or mountaintop weather sensors), physical capture of equipment is an operational threat. DART addresses data remanence through strict defensive measures:

- **Volatile Memory Zeroization:** All sensitive buffers—including raw passkeys, expanded Argon2 scratchpads, symmetric key buffers, and plaintext arrays—are sanitized using explicit memory wipes (`crypto_wipe()`), preventing compilers from optimizing away scrub routines via dead-store elimination.
- **File Shredding (`wipe_sent`):** When enabled, sent telemetry files are overwritten with zeros across physical sectors and flushed via `fsync()` before unlinking, thwarting recovery from solid-state flash wear-leveling strata.
- **Amateur Radio Note:** Under amateur radio regulatory frameworks (e.g., FCC Part 97.113), messages intended to obscure the meaning of telecommunications are prohibited. For amateur radio operations, the passkey is left unconfigured, running DART in native cleartext mode with full FEC, while commercial/educational experimental licenses employ the authenticated encryption layer.

6 Hardware Interfacing & System Deployment

Operating at the physical layer requires direct interaction with host hardware interfaces:

- **Push-to-Talk (PTT) Keying:** DART interfaces with transceiver control lines using serial port modem control lines (RTS / DTR pin assertion via `UNIX ioctl(fd, TIOCMSET, &status)`). A programmable pre-transmit delay ($T_{\text{txdelay}} \approx 100\text{--}250\text{ ms}$) accommodates relay stabilization.
- **Carrier Sense Multiple Access (CSMA):** Prior to keying the transmitter, the background listener daemon (dartw) samples the RF channel. If received audio energy exceeds `watch_squelch_dbfs` (e.g., -25 dBFS), the channel is marked busy, and transmission defers, preventing collisions on shared amateur frequencies.
- **Audio I/O:** Baseband waveforms are rendered and captured using PortAudio streams at native 48 kHz, 32-bit floating point (TX) or 16-bit integer PCM (RX).

7 Summary & Performance Verification

The DART standard demonstrates how low-level physical modulation and signal processing primitives can be implemented efficiently in portable C. By implementing 4-AFSK with 300 Hz orthogonal spacing, Gray coding, rate-1/2 constraint-7 convolutional FEC, dynamic matrix block interleaving, early/late symbol tracking, and soft-decision diversity combining, DART provides robust telemetry throughput within standard voice-grade amateur transceivers.

Parameter	Specification Value
Modulation Format	4-ary Audio Frequency Shift Keying (4-AFSK)
Center Frequency	1700 Hz
Tone Frequencies	1250 Hz, 1550 Hz, 1850 Hz, 2150 Hz
Symbol Rate	300 baud (Configurable)
Data Throughput	600 bits/s raw channel rate (300 bps net coded)
Bit Mapping	2-bit Reflected Gray Code
FEC Code	Rate 1/2 Convolutional, $K = 7$, $G_1 = 171_8$, $G_2 = 133_8$
Interleaver	Adaptive Matrix Square/Rectangular Block
Frame Synchronization	32-bit ASM (0x1ACFFC1D)
Integrity Check	CRC-8 (Header), CRC-CCITT 16 (Payload)
AEAD Encryption	XChaCha20-Poly1305 (24-byte Nonce, 16-byte MAC)
Key Derivation	Argon2id (3 passes, configurable memory, 128-bit salt)
Key Fingerprint	6-character hex via BLAKE2b ("DART-keyid-v1")
Channel Access	Squelch Carrier Sense Energy Detection (CSMA/CD)